



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

The Importance of Different Types of Models

Process Data Constraints Key Table **RELATIONAL** Tuple Relation **DATA MODEL** Storage Efficiency Instance Schema

Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to relational database management system in spoken or written texts.

Describe relational database, its concepts, and principles in spoken or written texts.

Can Do Statements:

- I can recall basic information about the relational data model in DBMS.
- I can discuss basic concepts in the relational data model.
- I can talk about constraints in the relational data model.

Ask your partner:

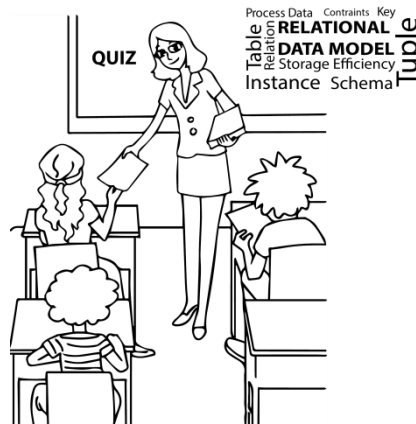
- How do you think a database can be related to another one?
- What types of relationships are there among databases?





Let's Explore!

Quick Oral Quiz



Instructions: Read the text below.

Teacher: Ok Mark, today I will test your knowledge; I will make some questions. The answers are very simple. The exam is as difficult as the last one.

Mark: Understood, sir.

Teacher: First, tell me about the relational data model.

Mark: It is the main data model used by almost everybody to store and process data. It is used often because it provides storage efficiency.

Teacher: Can you develop a little bit more on that? Are the relations saved as tables?

Mark: Yes sir. There are six concepts.

Tables are the ones that store all the relations between entities in form of charts with rows and columns. The Tuple is one row of a table and it has only one record. Then, there is the relation instance, which is a determined number of tuples; the relation schema that presents the associations between the table names, attributes and the attribute's names. The relation key is the attributes that identify a specific row exclusively. Finally, the attribute domain, which is the pre-assigned importance of an attribute.

Teacher: Excellent Matt! Now what are the requirements needed for a relation to be valid?

Matt: That is not as easy to explain as you might think. For a relation to be valid, there are some integrity relational constraints that need to be accomplished. There are three kinds: the key constraints, that is the slight subset of attributes that make a tuple unique; the domain constraints, which means that every attribute has to have a detailed set of values that cannot be unreal; and the referential integrity constraints, that refers to the connection that can exist between two key attributes in different relations.

Teacher: Good job, Matt! One last thing, can the relational model be taken as the primary data model?

Matt: Well, as long as it works for the specific purposes of a project, yes, sir!





Let's Practice!

Can data model information be easily recalled?

Instructions: Complete the sentences with the words from the box. Read the dialogue aloud. Check your pronunciation with your instruction.

data is processed and stored / can be said / is used and applied /
it may be stated / this model is well recognized

Joy: How much information can you recall about the relational data model?

Teddy: Not much, I'd say **it may be stated** that Relational Data Model is **used and applied** by many people, such as software programmers around the world.

Joy: Yes, Also remember that it **can be said** that It is the main data model that helps to be able to have the data stored. But, I have a doubt, is also data processed?

Teddy: Oh sure, **data is processed and stored** all the while. I remember now that **this model is well recognized** because of its storage efficiency.



Let's Practice!

Some important concepts that need to be taken into account when dealing with RDM.

Instructions: Complete the crossword.

Down

1. In RDM there are six concepts worth to be taken into account, one of them is c.....
2. The R Data Model is usually abbreviated with the acronym RDM for its better understanding.

Across

3. The relation i is a bunch of numbers of tuples.
4. The tables can be thought as a s..... of data and relations.
5. Rows and columns shape into t.....where the relation of the entities occur.
6. The t..... is easily identified as it is one row in the table with one record.





Extension Activity

I can recall everything about Relational Data Model

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Write a short summary about Relational Data Model.
- Create a poster with the main concepts of Relational Data Model.
- Share with your partners the post you created.

Self-Assessment

Yes

No

- I write a short summary about Relational Data Model.
- I create a post with the main concepts of Relational Data Model.
- I share with your partners the post I created.

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Target Words

- table** (noun): structure used to organize information
e.g. Then, there is the relation instance, which is a determined number of tuples.
- constraints** (noun): a limitation or restriction
e.g. There are some integrity relational constraints that need to be accomplished.
- integrity** (noun): something that is complete that is a whole
e.g. There are some integrity relational constraints that need to be accomplished.
- process** (noun): a series of actions leading to an objective
e.g. It is the main data model used by almost everybody to store and process data.
- relations** (noun): a connection or association between two or more parts
e.g. Tables are the ones that store all the relations between entities in form of charts with rows and columns.
- tuple** (noun): a row of values in a relational database
e.g. The Tuple is one row of a table and it has only one record.



Learning Points

Simple Present Passive Interrogative

Simple present passive interrogative is mainly used to give more importance to the action instead to the person performing it.

Simple present passive interrogative sometimes is used because it is more polite.

The structure for simple present passive interrogative is:

(WH question) Am/ Is / Are + object + verb past participle

Example:

- *How many concepts **are included** in the relational data model?*

Examples

- *Are the relations saved as tables?*
- *What are the requirements needed for a relation to be valid?*
- *One last thing, can the relational model be taken as the primary data model?*



Learning Points

Comparisons

There are certain structures used to make **comparisons** in English two of them as: as... as and not as... as.

Some of the most used comparative connector are **not as... as** and **as... as**
The comparative **not as... as** is used to show that something is superior compared to another thing using a negative structure.

let's check its structure and examples:

one object+ verb to be + not as + adjective + as + second object

Example:

*-Tables are not **as easy to understand as** tuple.*

The comparative as... as is used to put to things at the same level, to say they are equal.

let's check its structure and examples:

one object+ verb to be + as + adjective + as + second object

Example:

*The relation instance is **as difficult as** the relation schema.*

Examples

- *That is not as easy to explain as you might think.*
- *The exam is as difficult as the last one.*
- *Well, as long as it works for the specific purposes of a project, yes, sir!*

